

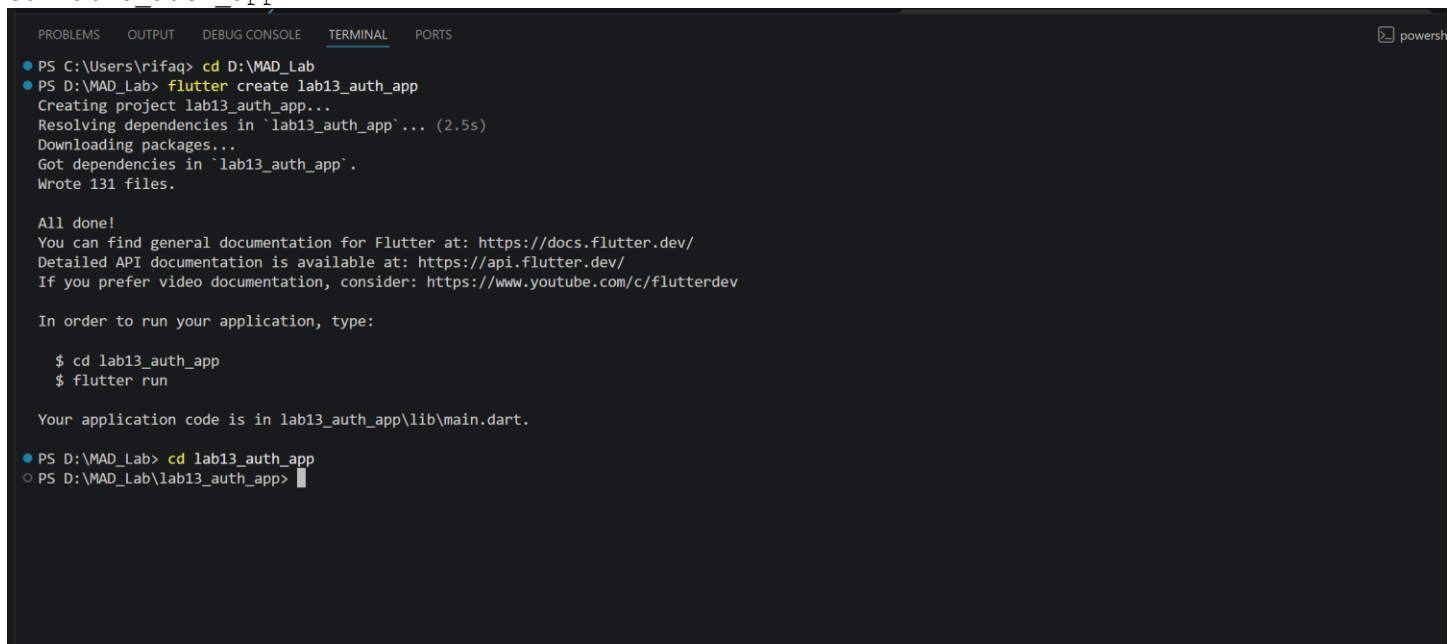
Task Requirement:

- Firebase Authentication
- Login Page
- Signup Page
- Validation
- Confirm Password
- Loader in button
- Disable button during loading
- Show/Hide password
- Success/Error snackbar
- Dashboard navigation

STEP 1 : Create Project

In terminal:

```
cd D:\MAD_Lab
flutter create lab13_auth_app
cd lab13_auth_app
```



The screenshot shows a Windows terminal window with the following output:

```
PS C:\Users\rifaq> cd D:\MAD_Lab
PS D:\MAD_Lab> flutter create lab13_auth_app
Creating project lab13_auth_app...
Resolving dependencies in 'lab13_auth_app'... (2.5s)
Downloading packages...
Got dependencies in 'lab13_auth_app'.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd lab13_auth_app
$ flutter run

Your application code is in lab13_auth_app\lib\main.dart.
PS D:\MAD_Lab> cd lab13_auth_app
PS D:\MAD_Lab\lab13_auth_app>
```

- Open project in Visual Studio Code

STEP 2 : Add Firebase Packages

pubspec.yaml

Inside dependencies, we will add:

```
firebase_core: ^3.13.0  
firebase_auth: ^5.5.0
```

```
29 # versions available, run `flutter pub outdated`  
30 dependencies:  
31 flutter:  
32   sdk: flutter  
33  
34   # The following adds the Cupertino Icons font to your application.  
35   # Use with the CupertinoIcons class for iOS style icons.  
36   cupertino_icons: ^1.0.8  
37  
38   firebase_core: ^3.13.0  
39   firebase_auth: ^5.5.0
```

STEP 3 : Install Packages

flutter pub get

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
● PS D:\MAD_Lab\lab13_auth_app> flutter pub get  
Resolving dependencies...  
Downloading packages...  
  matcher 0.12.19 (0.12.20 available)  
  meta 1.17.0 (1.18.2 available)  
  test_api 0.7.10 (0.7.12 available)  
  vector_math 2.2.0 (2.3.0 available)  
Got dependencies!  
4 packages have newer versions incompatible with dependency constraints.  
Try `flutter pub outdated` for more information.  
○ PS D:\MAD_Lab\lab13_auth_app> 
```

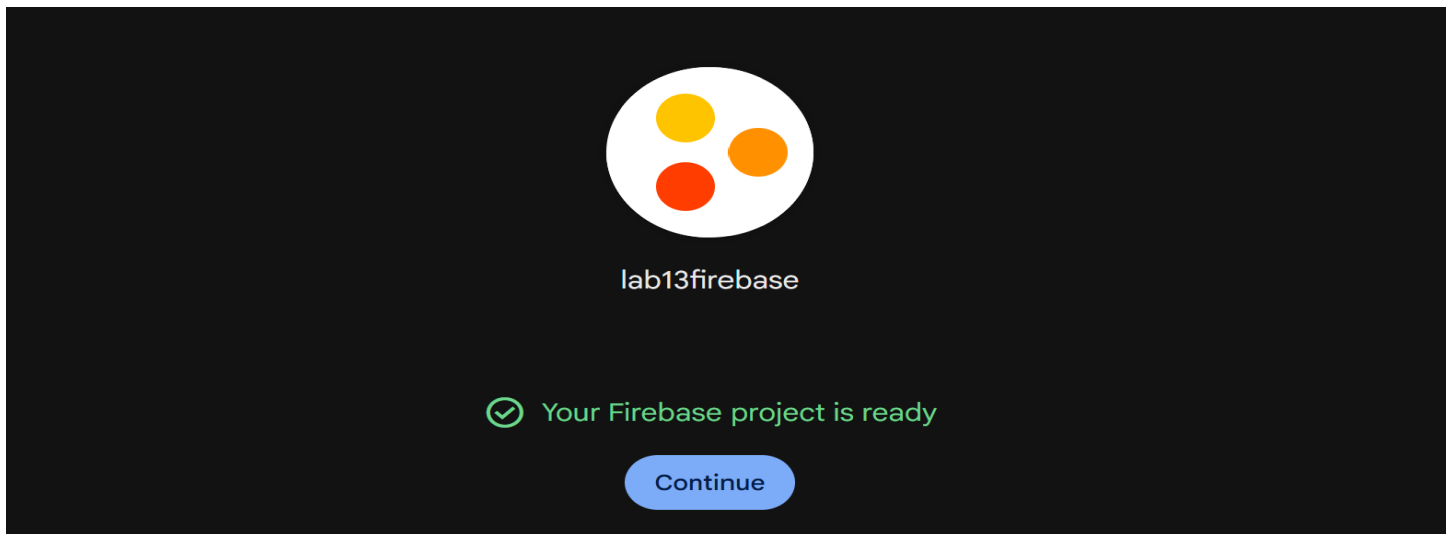
STEP 4 : Create Firebase Project

We will Open:

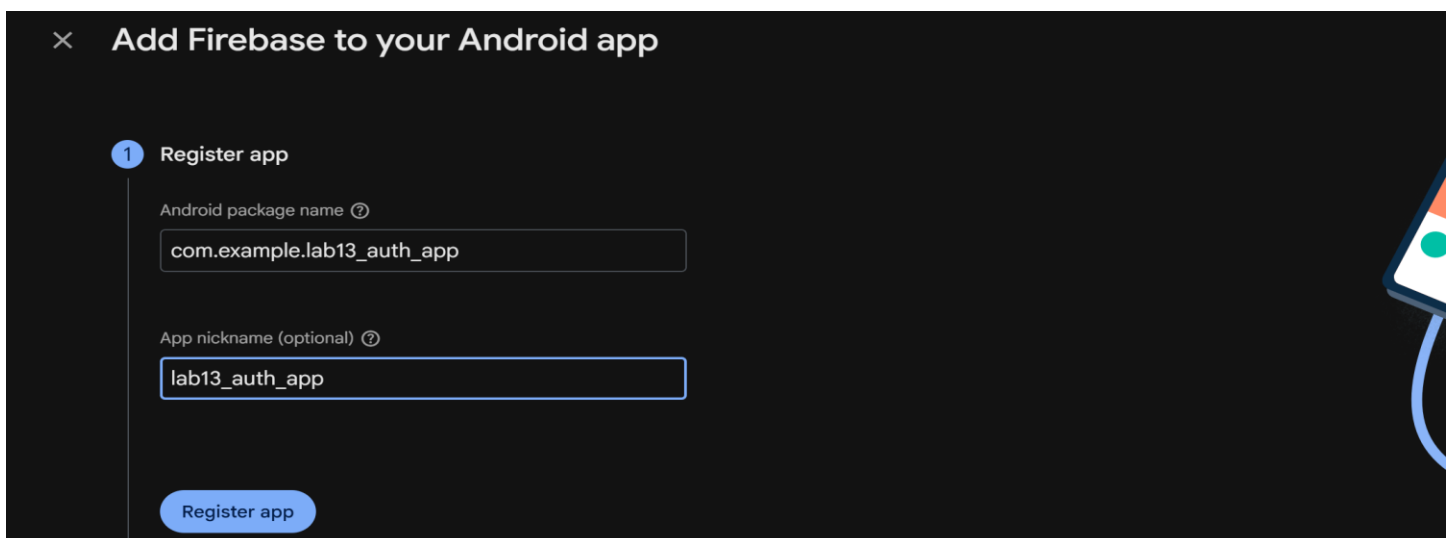
[Firebase Console](#)

Create project:

lab13firebase



STEP 5 — Add Android App + Download `google-services.json`

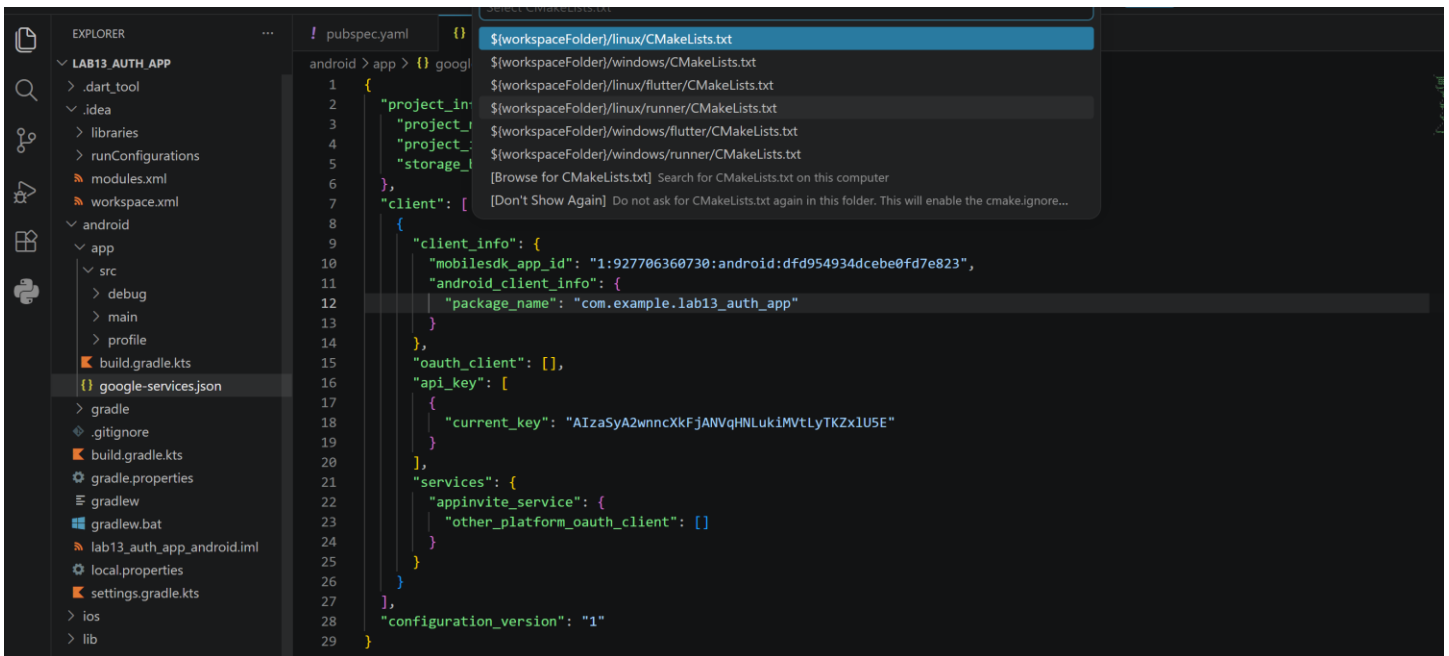


We will download google-services.json

Move File Into Flutter Project

google-services.json

```
lab13_auth_app/  
└─ android/  
    └─ app/  
        └─ google-services.json
```



STEP 6 : Configure FlutterFire

dart pub global activate flutterfire cli



```
PS D:\MAD_Lab\lab13_auth_app> dart pub global run flutterfire_cli:flutterfire configure
i Found 2 Firebase projects.
✓ Select a Firebase project to configure your Flutter application with · lab13firebase-a45e7 (lab13firebase)
✓ Which platforms should your configuration support (use arrow keys & space to select)? · android, ios, macos, web, windows
i Firebase android app com.example.lab13_auth_app registered.
i Firebase ios app com.example.lab13AuthApp is not registered on Firebase project lab13firebase-a45e7.
i Registered a new Firebase ios app on Firebase project lab13firebase-a45e7.
i Firebase macos app com.example.lab13AuthApp registered.
i Firebase web app lab13_auth_app (web) is not registered on Firebase project lab13firebase-a45e7.
i Registered a new Firebase web app on Firebase project lab13firebase-a45e7.
i Firebase windows app lab13_auth_app (windows) is not registered on Firebase project lab13firebase-a45e7.
i Registered a new Firebase windows app on Firebase project lab13firebase-a45e7.

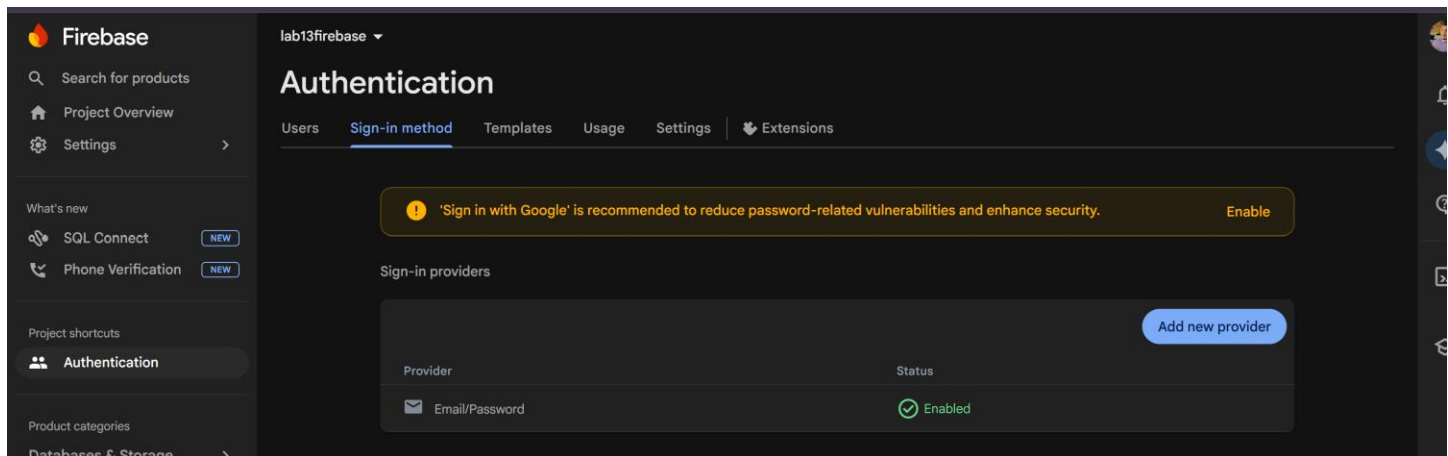
Firebase configuration file lib\firebase_options.dart generated successfully with the following Firebase apps:

Platform  Firebase App Id
web       1:927706360730:web:1f3ce4b5e996b3b7d7e823
android   1:927706360730:android:dfd954934dcebe0fd7e823
ios       1:927706360730:ios:9e02d281857c1c96d7e823
macos     1:927706360730:ios:9e02d281857c1c96d7e823
windows   1:927706360730:web:0e6e422e6479f535d7e823

Learn more about using this file and next steps from the documentation:
> https://firebase.google.com/docs/flutter/setup
PS D:\MAD_Lab\lab13_auth_app>
```

STEP 7: Enable Authentication

Email/password enable



STEP 8 : Create Screens Folder

lib/

We will Create folder:

screens

Inside it we will create 3 files:

login_screen.dart
signup_screen.dart
dashboard_screen.dart

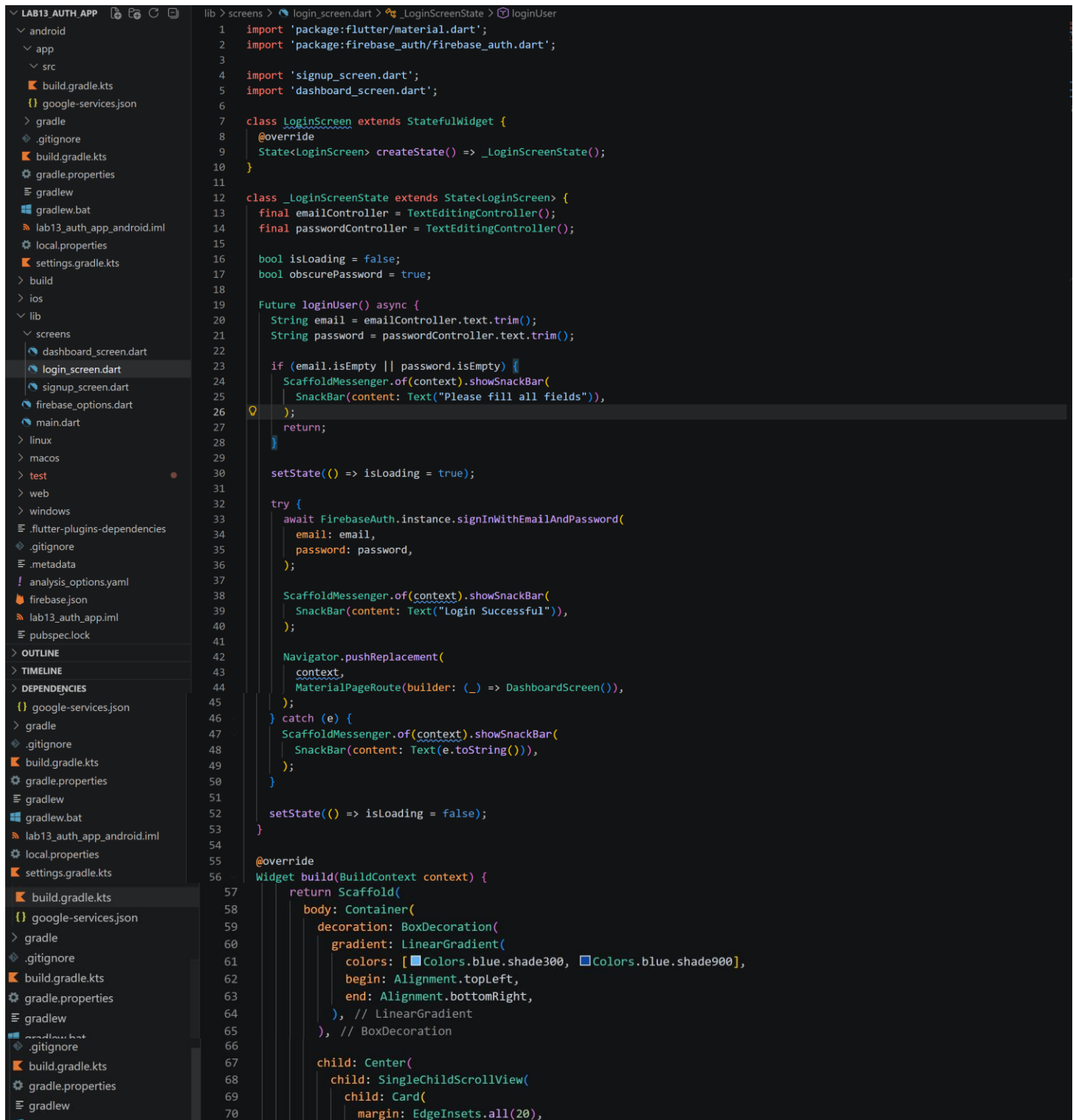
STEP 9 : Replace main.dart

lib/main.dart

```
lib > main.dart > MyApp > build
1  import 'package:flutter/material.dart';
2  import 'package:firebase_core/firebase_core.dart';
3  import 'firebase_options.dart';
4  import 'screens/login_screen.dart';
5
6  void main() async {
7
8      WidgetsFlutterBinding.ensureInitialized();
9
10     await Firebase.initializeApp(
11         | options: DefaultFirebaseOptions.currentPlatform,
12         | );
13
14     runApp(MyApp());
15 }
16
17 class MyApp extends StatelessWidget {
18
19     @override
20     Widget build(BuildContext context) {
21
22         return MaterialApp(
23             | debugShowCheckedModeBanner: false,
24             | home: LoginScreen(),
25             | );
26     }
27 }
```

STEP 10 : Login Screen

lib/screens/login_screen.dart



```

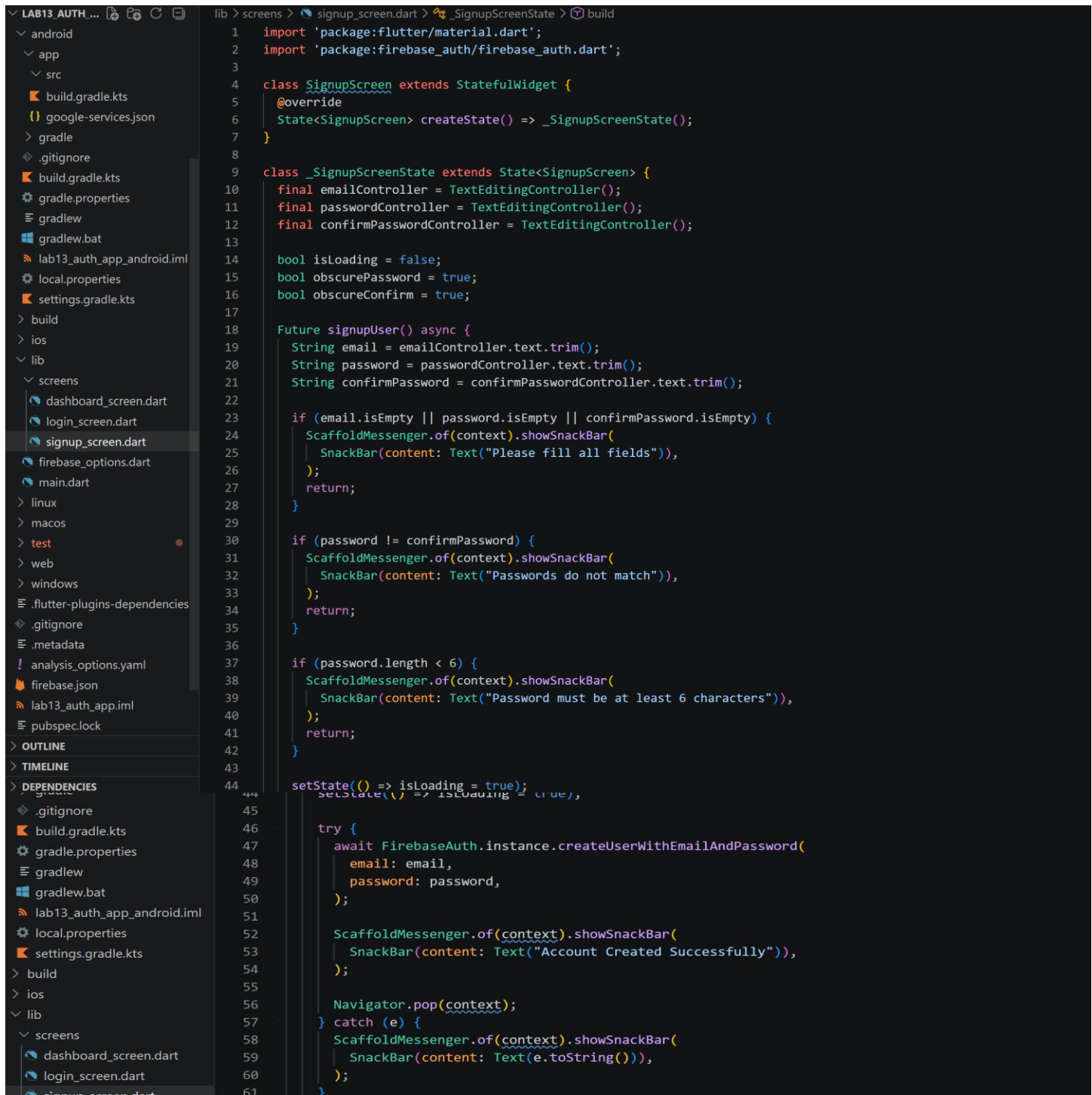
1  import 'package:flutter/material.dart';
2  import 'package:firebase_auth/firebase_auth.dart';
3
4  import 'signup_screen.dart';
5  import 'dashboard_screen.dart';
6
7  class LoginScreen extends StatefulWidget {
8    @override
9    State<LoginScreen> createState() => _LoginScreenState();
10 }
11
12 class _LoginScreenState extends State<LoginScreen> {
13   final emailController = TextEditingController();
14   final passwordController = TextEditingController();
15
16   bool isLoading = false;
17   bool obscurePassword = true;
18
19   Future loginUser() async {
20     String email = emailController.text.trim();
21     String password = passwordController.text.trim();
22
23     if (email.isEmpty || password.isEmpty) {
24       ScaffoldMessenger.of(context).showSnackBar(
25         SnackBar(content: Text("Please fill all fields")),
26       );
27     }
28     return;
29   }
30
31   setState(() => isLoading = true);
32
33   try {
34     await FirebaseAuth.instance.signInWithEmailAndPassword(
35       email: email,
36       password: password,
37     );
38     ScaffoldMessenger.of(context).showSnackBar(
39       SnackBar(content: Text("Login Successful")),
40     );
41     Navigator.pushReplacement(
42       context,
43       MaterialPageRoute(builder: (_) => DashboardScreen()),
44     );
45   } catch (e) {
46     ScaffoldMessenger.of(context).showSnackBar(
47       SnackBar(content: Text(e.toString())),
48     );
49   }
50
51   setState(() => isLoading = false);
52 }
53
54 @override
55 Widget build(BuildContext context) {
56   return Scaffold(
57     body: Container(
58       decoration: BoxDecoration(
59         gradient: LinearGradient(
60           colors: [Colors.blue.shade300, Colors.blue.shade900],
61           begin: Alignment.topLeft,
62           end: Alignment.bottomRight,
63         ), // LinearGradient
64       ), // BoxDecoration
65       child: Center(
66         child: SingleChildScrollView(
67           child: Card(
68             margin: EdgeInsets.all(20),

```

```
71 shape: RoundedRectangleBorder(  
72   borderRadius: BorderRadius.circular(20),  
73 ), // RoundedRectangleBorder  
74 elevation: 10,  
75  
76 child: Padding(  
77   padding: EdgeInsets.all(20),  
78  
79   child: Column(  
80     mainAxisAlignment: MainAxisAlignment.min,  
81     children: [  
82  
83       Text(  
84         "LOGIN",  
85         style: TextStyle(  
86           fontSize: 28,  
87           fontWeight: FontWeight.bold,  
88         ), // TextStyle  
89       ), // Text  
90  
91       SizedBox(height: 20),  
92  
93       TextField(  
94         controller: emailController,  
95         decoration: InputDecoration(  
96           prefixIcon: Icon(Icons.email),  
97           labelText: "Email",  
98           border: OutlineInputBorder(  
99             borderRadius: BorderRadius.circular(12),  
100           ), // OutlineInputBorder  
101         ), // InputDecoration  
102       ), // TextField  
103  
104       SizedBox(height: 15),  
105  
106       TextField(  
107         controller: passwordController,  
108         obscureText: obscurePassword,  
109         decoration: InputDecoration(  
110           prefixIcon: Icon(Icons.lock),  
111           labelText: "Password",  
112           border: OutlineInputBorder(  
113             borderRadius: BorderRadius.circular(12),  
114           ), // OutlineInputBorder  
115           suffixIcon: IconButton(  
116             icon: Icon(  
117               obscurePassword  
118                 ? Icons.visibility  
119                 : Icons.visibility_off,  
120             ), // Icon  
121             onPressed: () {  
122               setState(() {  
123                 obscurePassword = !obscurePassword;  
124               });  
125             }, // IconButton  
126           ), // InputDecoration  
127       ), // TextField  
128  
129       SizedBox(height: 20),  
130  
131       SizedBox(  
132         width: double.infinity,  
133         height: 50,  
134         child: ElevatedButton(  
135           onPressed: isLoading ? null : loginUser,  
136           child: isLoading  
137             ? CircularProgressIndicator(color: Colors.white)  
138             : Text("Login"),  
139         ), // ElevatedButton  
140       ), // SizedBox  
141  
142       TextButton(  
143         onPressed: () {  
144           Navigator.push(  
145             context,  
146             MaterialPageRoute(builder: (_) => SignupScreen()),  
147           );  
148         },  
149         child: Text("Don't have an account? Sign Up"),  
150       ), // TextButton  
151     ],  
152   ), // Column  
153 ), // Padding  
154 ), // Card  
155 ), // SingleChildScrollView  
156 ), // Center  
157 ), // Container  
158 ); // Scaffold  
159 }  
160 }  
161 }
```


STEP 11 — SIGNUP SCREEN

lib/screens/signup_screen.dart



```
1 import 'package:flutter/material.dart';
2 import 'package:firebase_auth/firebase_auth.dart';
3
4 class SignupScreen extends StatefulWidget {
5   @override
6   State<SignupScreen> createState() => _SignupScreenState();
7 }
8
9 class _SignupScreenState extends State<SignupScreen> {
10   final emailController = TextEditingController();
11   final passwordController = TextEditingController();
12   final confirmPasswordController = TextEditingController();
13
14   bool isLoading = false;
15   bool obscurePassword = true;
16   bool obscureConfirm = true;
17
18   Future signupUser() async {
19     String email = emailController.text.trim();
20     String password = passwordController.text.trim();
21     String confirmPassword = confirmPasswordController.text.trim();
22
23     if (email.isEmpty || password.isEmpty || confirmPassword.isEmpty) {
24       ScaffoldMessenger.of(context).showSnackBar(
25         SnackBar(content: Text("Please fill all fields")),
26       );
27       return;
28     }
29
30     if (password != confirmPassword) {
31       ScaffoldMessenger.of(context).showSnackBar(
32         SnackBar(content: Text("Passwords do not match")),
33       );
34       return;
35     }
36
37     if (password.length < 6) {
38       ScaffoldMessenger.of(context).showSnackBar(
39         SnackBar(content: Text("Password must be at least 6 characters")),
40       );
41       return;
42     }
43
44     setState(() => isLoading = true);
45     try {
46       await FirebaseAuth.instance.createUserWithEmailAndPassword(
47         email: email,
48         password: password,
49       );
50
51       ScaffoldMessenger.of(context).showSnackBar(
52         SnackBar(content: Text("Account Created Successfully")),
53       );
54
55       Navigator.pop(context);
56     } catch (e) {
57       ScaffoldMessenger.of(context).showSnackBar(
58         SnackBar(content: Text(e.toString())),
59       );
60     }
61   }
```

```

62 build.gradle.kts
63 {} google-services.json
64
65 > gradle
66
67 build.gradle.kts
68 gradle.properties
69 gradlew
70 gradlew.bat
71 lab13_auth_app_android.iml
72 local.properties
73 settings.gradle.kts
74
75 build
76 ios
77 lib
78 screens
79 dashboard_screen.dart
80 login_screen.dart
81 signup_screen.dart
82 firebase_options.dart
83 main.dart
84
85 linux
86 macos
87 test
88 web
89 windows
90 .flutter-plugins-dependencies
91 .gitignore
92 .metadata
93 analysis_options.yaml
94 firebase.json
95 lab13_auth_app.iml
96 pubspec.lock
97
98 OUTLINE
99
100 TIMELINE
101
102 DEPENDENCIES
103 > gradle
104 .gitignore
105 build.gradle.kts
106 gradle.properties
107 gradlew
108 gradlew.bat
109 lab13_auth_app_android.iml
110 local.properties
111 settings.gradle.kts
112 build
113 ios
114 lib
115 screens
116 dashboard_screen.dart
117 login_screen.dart
118 signup_screen.dart
119 firebase_options.dart
120 main.dart
121 linux
122 macos
123 test
124 web
125 windows
126 .flutter-plugins-dependencies
127 .gitignore
128 .metadata
129 analysis_options.yaml
130 firebase.json
131 lab13_auth_app.iml
132 pubspec.lock
133
134 OUTLINE
135 TIMELINE
136 DEPENDENCIES

```

```

63 setState(() => isLoading = false);
64 }
65
66 @override
67 Widget build(BuildContext context) {
68   return Scaffold(
69     body: Container(
70       decoration: BoxDecoration(
71         gradient: LinearGradient(
72           colors: [Colors.purple.shade300, Colors.purple.shade900],
73           begin: Alignment.topLeft,
74           end: Alignment.bottomRight,
75         ), // LinearGradient
76       ), // BoxDecoration
77
78       child: Center(
79         child: SingleChildScrollView(
80           child: Card(
81             margin: EdgeInsets.all(20),
82             shape: RoundedRectangleBorder(
83               borderRadius: BorderRadius.circular(20),
84             ), // RoundedRectangleBorder
85             elevation: 10,
86
87             child: Padding(
88               padding: EdgeInsets.all(20),
89
90               child: Column(
91                 mainAxisAlignment: MainAxisAlignment.min,
92                 children: [
93                   Text(
94                     "SIGN UP",
95                     style: TextStyle(
96                       fontSize: 28,
97                       fontWeight: FontWeight.bold,
98                     ), // TextStyle
99                   ), // Text
100
101                   SizedBox(height: 20),
102                   SizedBox(height: 20),
103                   TextField(
104                     controller: emailController,
105                     decoration: InputDecoration(
106                       prefixIcon: Icon(Icons.email),
107                       labelText: "Email",
108                       border: OutlineInputBorder(
109                         borderRadius: BorderRadius.circular(12),
110                       ), // OutlineInputBorder
111                     ), // InputDecoration
112                   ), // TextField
113
114                   SizedBox(height: 15),
115                   TextField(
116                     controller: passwordController,
117                     obscureText: obscurePassword,
118                     decoration: InputDecoration(
119                       prefixIcon: Icon(Icons.lock),
120                       labelText: "Password",
121                       border: OutlineInputBorder(
122                         borderRadius: BorderRadius.circular(12),
123                       ), // OutlineInputBorder
124                       suffixIcon: IconButton(
125                         icon: Icon(
126                           obscurePassword
127                             ? Icons.visibility
128                             : Icons.visibility_off,
129                         ), // Icon
130                         onPressed: () {
131                           setState(() {
132                             obscurePassword = !obscurePassword;
133                           });
134                         }, // IconButton
135                       ), // InputDecoration
136                     ), // TextField

```

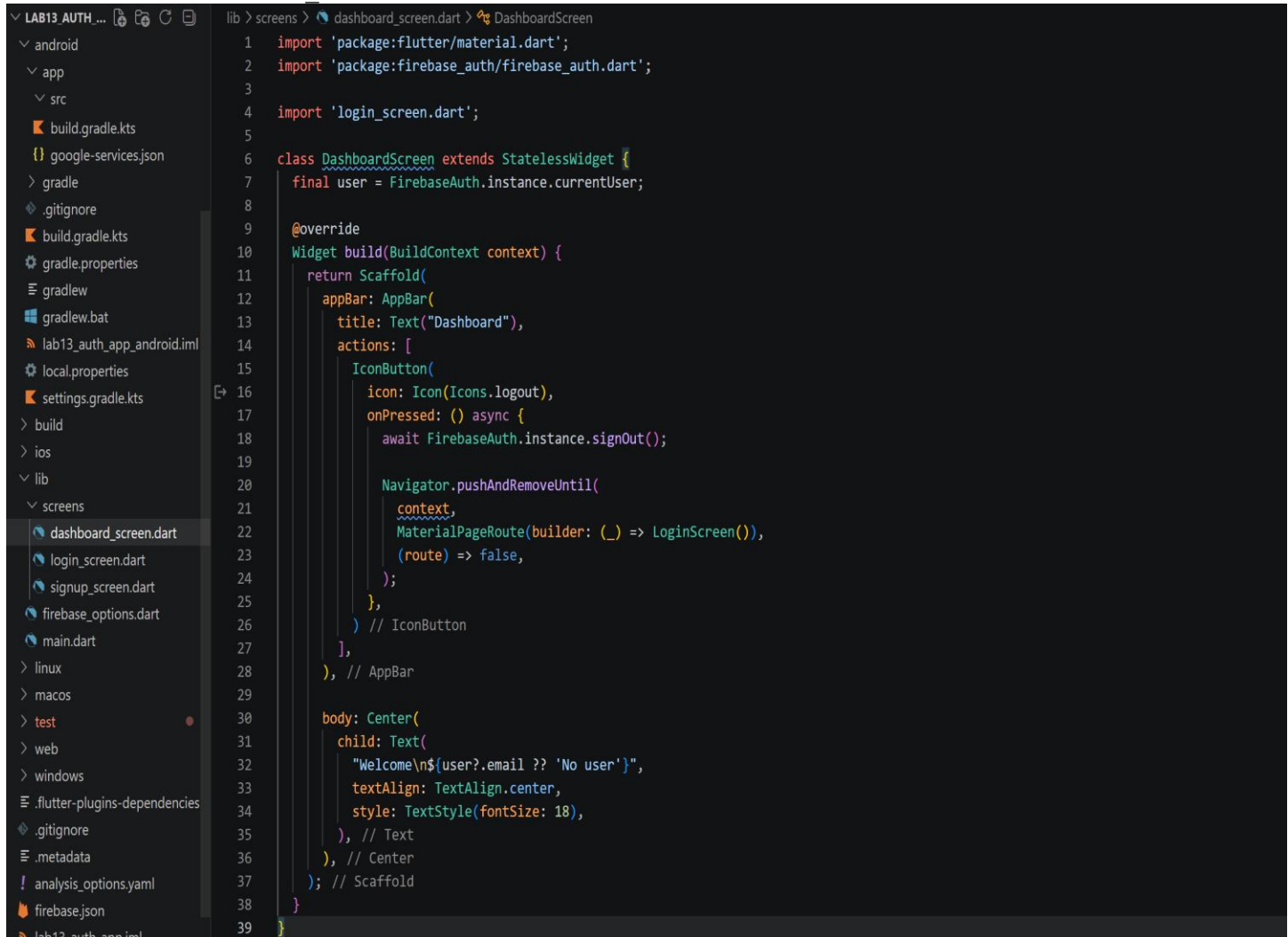
```
140 {} google-services.json
141 > gradle
142 .gitignore
143 build.gradle.kts
144 gradle.properties
145 gradlew
146 gradlew.bat
147 lab13_auth_app_android.iml
148 local.properties
149 settings.gradle.kts
150 build
151 ios
152 lib
153 screens
154   dashboard_screen.dart
155   login_screen.dart
156   signup_screen.dart
157   firebase_options.dart
158   main.dart
159 linux
160 macos
161 test
162 web
163 windows
164 .flutter-plugins-dependencies
165 .gitignore
166 .metadata
167 analysis_options.yaml
168 firebase.json
169 lab13_auth_app.iml
170 pubspec.lock
171 OUTLINE
172 TIMELINE
173 DEPENDENCIES

140 SizedBox(height: 15),
141
142 TextField(
143   controller: confirmPasswordController,
144   obscureText: obscureConfirm,
145   decoration: InputDecoration(
146     prefixIcon: Icon(Icons.lock_outline),
147     labelText: "Confirm Password",
148     border: OutlineInputBorder(
149       borderRadius: BorderRadius.circular(12),
150     ), // OutlineInputBorder
151     suffixIcon: IconButton(
152       icon: Icon(
153         obscureConfirm
154           ? Icons.visibility
155           : Icons.visibility_off,
156       ), // Icon
157       onPressed: () {
158         setState(() {
159           obscureConfirm = !obscureConfirm;
160         });
161       },
162     ), // IconButton
163   ), // InputDecoration
164 ), // TextField
165
166 SizedBox(height: 20),
167
168 SizedBox(
169   width: double.infinity,
170   height: 50,
171   child: ElevatedButton(
172     onPressed: isLoading ? null : signupUser,
173     child: isLoading
174       ? CircularProgressIndicator(color: Colors.white)
175       : Text("Sign Up"),
176   ), // ElevatedButton
177 ), // SizedBox
```

```
178   ), // SizedBox
179 ],
180 ), // Column
181 ), // Padding
182 ), // Card
183 ), // SingleChildScrollView
184 ), // Center
185 ), // Container
186 ); // Scaffold
187 }
188 }
```

STEP 12 : DASHBOARD SCREEN (FINAL PAGE)

lib/screens/dashboard screen.dart



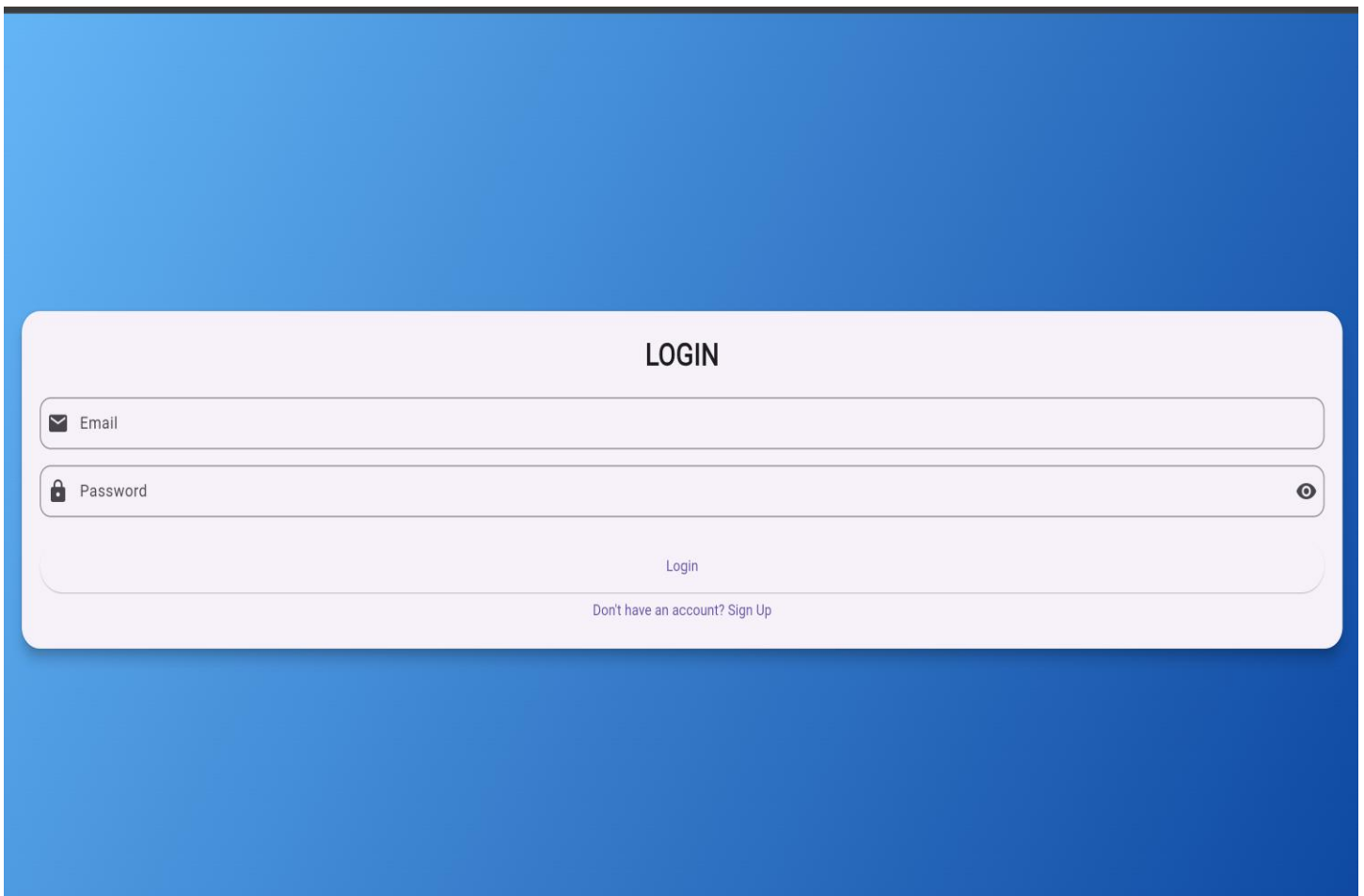
```
1  import 'package:flutter/material.dart';
2  import 'package:firebase_auth/firebase_auth.dart';
3
4  import 'login_screen.dart';
5
6  class DashboardScreen extends StatelessWidget {
7    final user = FirebaseAuth.instance.currentUser;
8
9    @override
10   Widget build(BuildContext context) {
11     return Scaffold(
12       appBar: AppBar(
13         title: Text("Dashboard"),
14         actions: [
15           IconButton(
16             icon: Icon(Icons.logout),
17             onPressed: () async {
18               await FirebaseAuth.instance.signOut();
19
20               Navigator.pushAndRemoveUntil(
21                 context,
22                 MaterialPageRoute(builder: (_) => LoginScreen()),
23                 (route) => false,
24               );
25             },
26           ) // IconButton
27         ],
28       ), // AppBar
29
30       body: Center(
31         child: Text(
32           "Welcome\n${user?.email ?? 'No user'}",
33           textAlign: TextAlign.center,
34           style: TextStyle(fontSize: 18),
35         ), // Text
36       ), // Center
37     ); // Scaffold
38   }
39 }
```

FINAL STEP : RUN PROJECT

```
flutter pub get
```

```
flutter run -d chrome
```

1. Login Screen



2. Signup Screen

SIGN UP

Email

Password

Confirm Password

Sign Up

SIGN UP

Email
rifaqajmalmohmand@gmail.com

Password
Rifaq123

Confirm Password
Rifaq123

Sign Up

LOGIN

Email
✉ guddagaming06@gmail.com

Password
🔒 @Rifaq@Uetian

Login

[Don't have an account? Sign Up](#)

Account Created Successfully

3. Dashboard Screen (After Login)

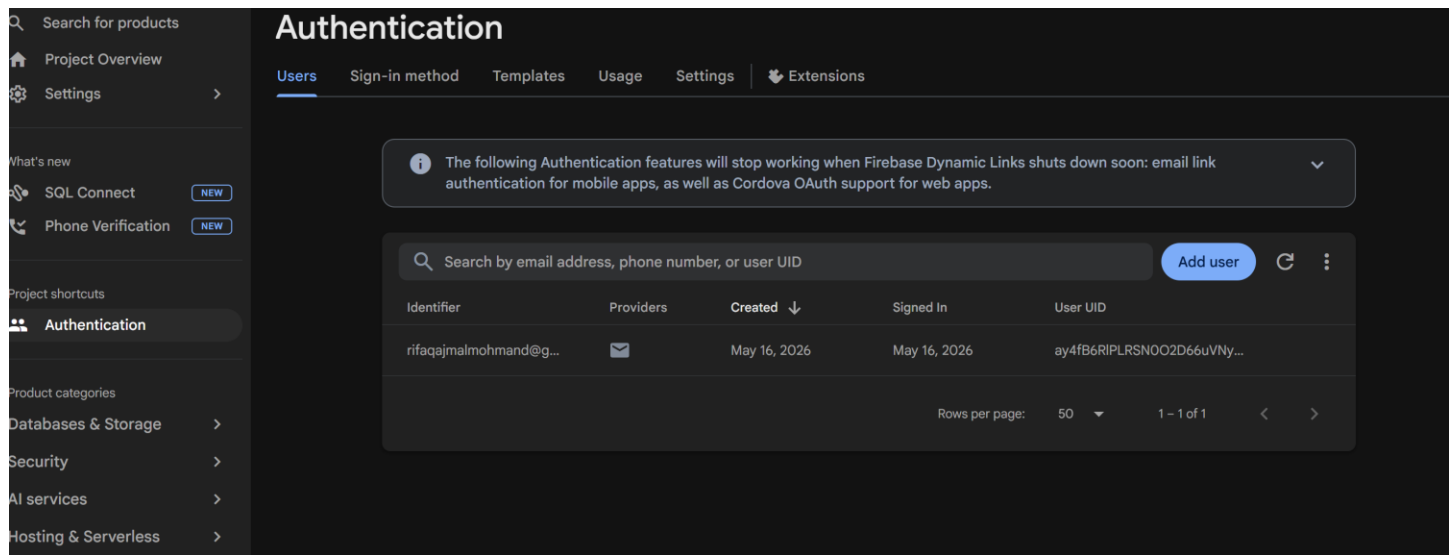
Dashboard ↻

Welcome
rifaqajmalmohmand@gmail.com

4. Firebase Authentication Console

[Firebase Console](#)

- Build → Authentication → Users tab
- registered user email



The screenshot displays the Firebase Authentication console interface. On the left is a sidebar with navigation options: Search for products, Project Overview, Settings, What's new (SQL Connect, Phone Verification), Project shortcuts (Authentication), and Product categories (Databases & Storage, Security, AI services, Hosting & Serverless). The main panel is titled 'Authentication' and has tabs for Users, Sign-in method, Templates, Usage, Settings, and Extensions. The 'Users' tab is active, showing a message about Firebase Dynamic Links deprecation and a search bar. Below the search bar is a table of users.

Identifier	Providers	Created ↓	Signed In	User UID
rifaqajmalmohmand@g...		May 16, 2026	May 16, 2026	ay4fB6RIPLRSN002D66uVNy...

At the bottom of the table, it shows 'Rows per page: 50' and '1 - 1 of 1'.